Oat Variety trial 2026

2026

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2026-11-06

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link to data:https://docs.google.com/spreadsheets/d/1-dI9AvKNXQALnqeG1BHwhA_7TwHpZvqA6yte4kpffxY/edit?gid=71330303#gid=71330303

1 Summary

1.1 Narrative

This trial was planted to gain familiarity with the oat/barley lab equipment. Trial was planted in the oat preliminary yield trial area on st paul field F1 on 29Apr2026 using the 10 row almaco drill which seeds 6 rows of a packet and the remaining 4 are winter wheat fill.

We planted 6 row plots which result in 3.5 ft x 12 ft plots, then trimmed them to 3.5 ft x 10 ft. The barley lab cultivated the plots during emergence but did not handweed our variety trial. As a result, the trial got weedy because we did not maintain the trial during the fieldseason. We did one handweeding event right before harvest to remove very large weeds (~2 hrs of work with 3 people). We then combine harvested the plots on August 5th.

1.2 Questions

How do we plant one of these trials?

How do we manage them for weeds?

How do we harvest?

1.3 Findings

Planting: Planting went well. We just need to plant on population rather than weight basis.

Weed management: We need handweeding at least twice in season. Starting in early June when plants get too big to cultivate.

Harvest: Using the combine scale is just as good as weighing in lab. Planting 6 rows in a 10 ft plot will not have enough sample for moisture and test weight at harvest in low yielding plots. Streaker is a hulless oat variety and needs a special correction for comparisons with hulled varieties when comparing yields in bushel per acre across varieties. This correction has not been applied yet.

Yield data: Of 24 plots, only 20 ended up making it to yield analysis due to issues with the data. We averaged around 20 bushels per acre which is very low, we expect around 80 bushels per acre. Moisture at harvest ranged from 10-12.5% which is acceptable.

1.4 photos

5	B		B		B		B		B		B		B		B	
5	B		401		402		403		404		405		406		B	
5	B		301		302		303		304		305		306		B	
5	B		201		202		203		204		205		206		B	
5	B		101		102		103		104		105		106		B	
5	B		B		B		B		B		B		B		B	
ft	10	5	10	5	10	5	10	5	10	5	10	5	10	5	10	
			sumo		rushmore		buffalo		streaker		reins		pearl			
			pearl		buffalo		streaker		sumo		reins		rushmore			
			buffalo		streaker		rushmore		sumo		reins		pearl			
			sumo		pearl		rushmore		streaker		reins		buffalo			

Figure 1: Variety trial plot map



Figure 2: Plots on 11June2026



Figure 3: Combine harvest 5Aug



Figure 4: Sample quality 5Aug2026

1.5 specs

Seed packaged in 2024, 1 gram of seed per sq ft, 45 grams per 3.5 x 12 ft seeded area

Planted 4/29/2026

Harvested 8/5/2026

Combine settings: drum=1540, distance=11mm, blower=3000

1.6 error acknowledgements

Plots not planted on a population basis

Trial very weedy

Sample sizes too small to compute test weight and moisture for all samples (20/24 had test weight) (16/24 had moisture)

2 Data QAQC

2.1 Yield HarvestMaster

Yield estimates using combine system (which we trust based on canola data)

Average MN Oat yield is 80 bu A. Over 100 is good, below 70 is low. below 25 is very low.

```
library(tidyverse)

read.csv("master_v_oat_26 - final.csv") -> dat
```

```
dat |>
  rename(yield_combine32TW_buA = yield_combine32TW..bu.A.,
         yield_combineAdjTW_buA = yield_combineAdjTW..bu.A.,
         yield_hand32TW_buA = yield_hand32TW..bu.A.,
         yield_handAdjTW_buA = yield_handAdjTW..bu.A.)  -> dat
```

```
dat |>
  arrange(plot) |>
  glimpse()
```

```
Rows: 48
Columns: 10
$ plot                      <chr> "101", "102", "103", "104", "105", "106", "~
$ variety                   <chr> "sumo", "pearl", "rushmore", "streaker", "r~
$ yield_combine32TW_buA     <dbl> 17.3841502, 34.5931853, -0.3187613, 9.03398~
$ yield_combineAdjTW_buA    <dbl> 22.48423, NA, NA, 13.99631, NA, 16.98863, 1~
$ yield_hand32TW_buA        <dbl> 30.1119808, 1.2507395, 0.6510699, 16.833582~
$ yield_handAdjTW_buA       <dbl> 30.39695, NA, NA, 18.96742, NA, 16.45580, 1~
$ lodgingscore_10flat_0erect <int> 6, 10, 10, 10, 6, 10, 9, 3, 10, 9, 8, 10, 8~
$ notes                     <chr> "very dirty", "", "", "very dirty", "empty"~
$ moisture_perten5200       <dbl> NA, NA, NA, NA, NA, 9.9, 10.7, NA, 10.2, 12~
$ testweight_perten5200     <dbl> 31.7, NA, NA, 28.4, NA, 35.5, 26.3, NA, 24.~
```

There are 4 estimates of yield

- yield_combine32TW = yield where weight is collected by combine and test weight is set to 32 lb per bu
- yield_combineAdjTW = yield where weight data is from combine and test weight collected on a perten 5200 is used. Many samples did not have sufficient volume for this calculatoin.

- yield_hand32TW = yield where weight data comes from subsample in combine which should be entire combine sample but may differ and test weight is set to 32
- yield_handAdjTW = yield where weight data comes from subsample in combine and test weight adjusted based on Perten 5200.

```
dat |>
  filter(variety!="b") |>
  select(plot, notes, testweight_perten5200, moisture_perten5200, yield_combine32TW_buA)
```

```
   plot               notes testweight_perten5200 moisture_perten5200
1   101          very dirty                  31.7                   NA
2   102                                       NA                   NA
3   103                                       NA                   NA
4   104          very dirty                  28.4                   NA
5   105               empty                    NA                   NA
6   106                                       35.5                  9.9
7   201                                       26.3                 10.7
8   202                                       NA                   NA
9   203                                       24.3                 10.2
10  204                                       17.4                 12.5
11  205                                       37.0                 10.4
12  206                                       30.5                 10.0
13  301                                       31.6                 10.0
14  302                                       31.9                 10.1
15  303                                       22.3                   NA
16  304                                       37.4                 10.9
17  305                                       36.5                 10.6
18  306 pratially mowed off                   36.5                 10.2
19  401                                       39.9                 11.1
20  402                                       34.6                 10.5
21  403                                       30.3                 10.0
22  404                                       15.6                   NA
23  405                                       36.4                 10.5
24  406                                       31.1                 10.0
   yield_combine32TW_buA
1             17.3841502
2             34.5931853
3             -0.3187613
4              9.0339836
5              0.0000000
6             13.1107889
```

```
7              8.3918778
8              1.3706734
9              7.5252656
10             4.4810476
11            28.9651733
12            22.5907279
13            18.5161141
14            31.4309585
15             5.8079380
16            23.3014475
17            32.1321792
18            26.5291622
19            30.3656872
20            27.2057898
21            29.4592488
22             3.8298227
23            28.2080608
24            16.6438910
```

```
#
# dat |>
#   filter(variety != "b") |>
#   arrange(moisture_perten5200) |>
#   select(plot, moisture_perten5200)
```

2.2 Outliers

```
dat |>
  filter(yield_combine32TW_buA<5) |>
  select(plot, variety, notes, yield_combine32TW_buA)
```

```
  plot  variety notes yield_combine32TW_buA
1    b        b                  0.0000000
2  103 rushmore                 -0.3187613
3  105    reins empty            0.0000000
4    b        b                  2.3330611
5  202 streaker                  1.3706734
6  204     sumo                  4.4810476
7  404 streaker                  3.8298227
```

There are two plots with no yield or negative yield. This is combine error and will be removed

```
dat |>
  filter(yield_combine32TW_buA>1)-> dat2
```

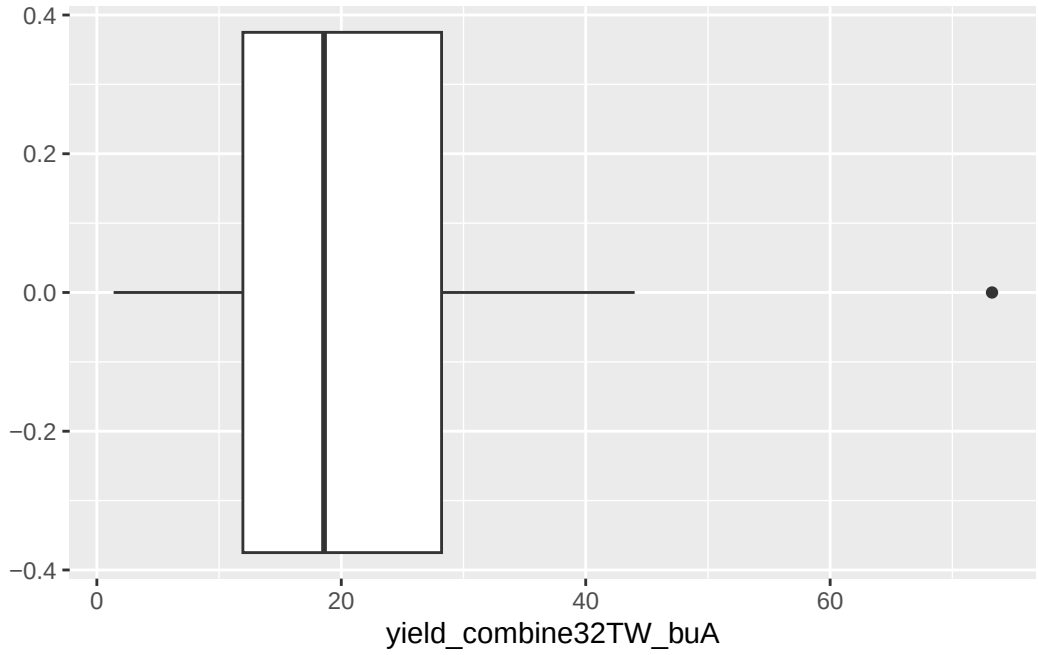
What about other plots in block 1

```
dat2 |>
  filter(plot<200) |>
  select(plot, variety, notes, yield_combine32TW_buA)
```

```
  plot  variety      notes yield_combine32TW_buA
1  101     sumo very dirty             17.384150
2  102    pearl                        34.593185
3  104 streaker very dirty              9.033984
4  106  buffalo                        13.110789
```

remainder of block 1 looks fine

```
dat2 |>
  ggplot(aes(yield_combine32TW_buA)) +
  geom_boxplot()
```



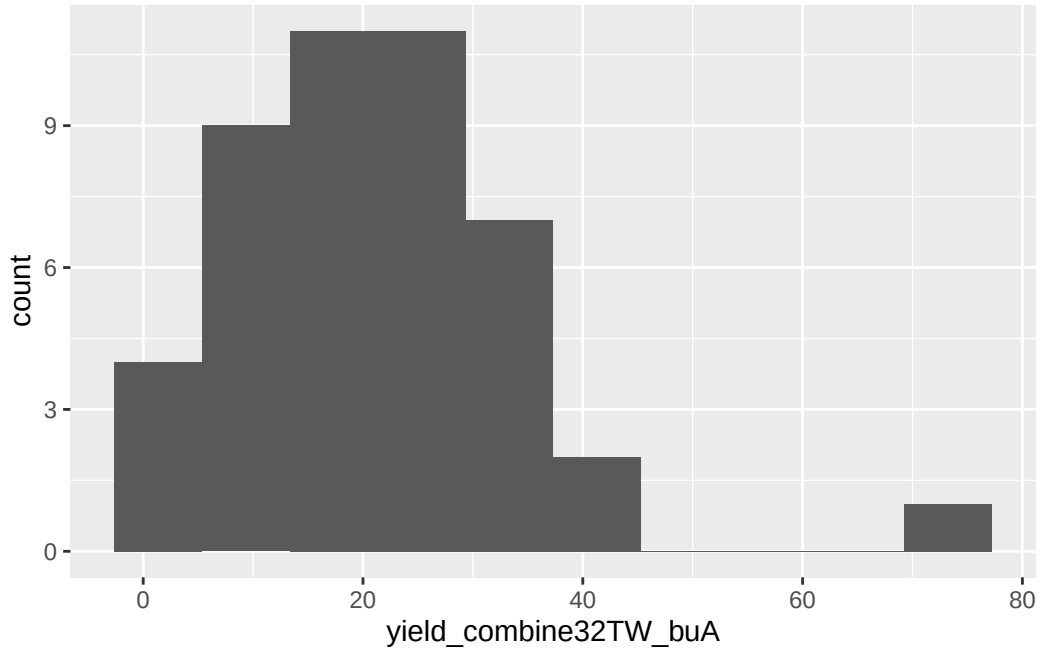
What about high yielding plot?

```
dat2 |>
  filter(yield_combine32TW_buA>50) |>
  select(plot, variety, yield_combine32TW_buA)
```

```
  plot variety yield_combine32TW_buA
1    b       b              73.24536
```

It's a border with relatively high yield

```
dat2 |>
  ggplot(aes(yield_combine32TW_buA)) +
  geom_histogram(bins = 10)
```



- Yields average around 20-30 bushels an acre. This is very low, but normally distributed
- I wonder if the 70 buA border was just 2 plots combined. I'm going to take out that plot
- The 100s was noted in combine as a messed up pass, it was the 300s and 400s where we started doing well

```
# taking out the 70 bu A plot. I think that a double measurement

dat2 |>
  filter(yield_combine32TW_buA<70) -> dat2
```

What is up with the 1 bu A plot in the 32TW estimates?

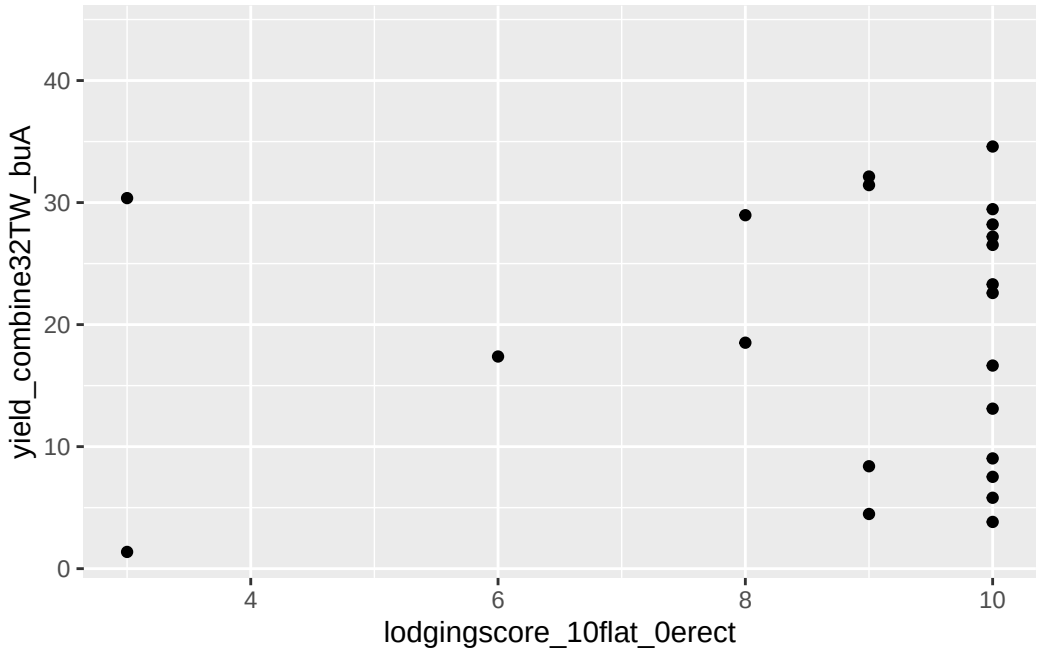
```
dat2 |>
  filter(yield_combine32TW_buA<8)
```

```
  plot  variety yield_combine32TW_buA yield_combineAdjTW_buA yield_hand32TW_buA
1    b        b              2.333061                     NA                  NA
2  202 streaker              1.370673                     NA            2.261611
3  203 rushmore              7.525266               15.79377           11.445123
4  204     sumo              4.481048               13.39164            8.301141
5  303 streaker              5.807938               15.36629           10.545618
6  404 streaker              3.829823               16.69410            7.315969
  yield_handAdjTW_buA lodgingscore_10flat_0erect notes moisture_perten5200
1                  NA                         NA                       NA
2                  NA                          3                       NA
3            15.07177                         10                     10.2
4            15.26647                          9                     12.5
5            15.13273                         10                       NA
6            15.00712                         10                       NA
  testweight_perten5200
1                    NA
2                    NA
3                  24.3
4                  17.4
5                  22.3
6                  15.6
```

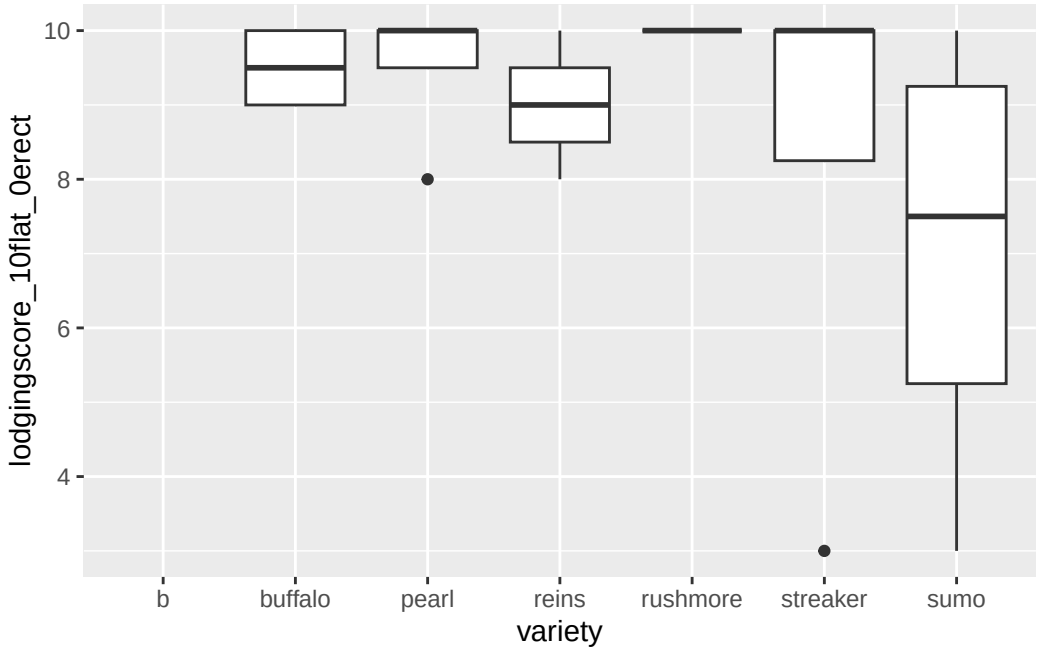
- 32TW plots are lower yielding than AdjTW plots because of the very low test weights
- combine and hand seem to track similarly for AdjTW
- Very low yielding plots unsurprisingly do not have TW data

2.3 Lodging

```
dat2 |>
  ggplot(aes(lodgingscore_10flat_0erect, yield_combine32TW_buA)) +
  geom_point()
```




```
dat2 |>
  ggplot(aes(variety, lodgingscore_10flat_0erect)) +
  geom_boxplot()
```



- Lots of lodging
- Sumo seems to have lower lodging
- No obvious trend between lodging and yield so not really going to investigate further

```
dat2 |>
  summary()
```

```
    plot             variety          yield_combine32TW_buA
Length:44          Length:44          Min.   : 1.371
Class :character   Class :character   1st Qu.:11.410
Mode  :character   Mode  :character   Median :18.566
                                      Mean   :19.726
                                      3rd Qu.:27.938
                                      Max.   :44.006

yield_combineAdjTW_buA yield_hand32TW_buA yield_handAdjTW_buA
Min.   :13.39          Min.   : 1.251     Min.   :15.01
1st Qu.:16.47          1st Qu.:11.711     1st Qu.:16.16
Median :37.42          Median :33.500     Median :36.25
Mean   :34.80          Mean   :32.567     Mean   :34.82
3rd Qu.:50.07          3rd Qu.:53.730     3rd Qu.:49.17
```

```
 Max.   :65.14          Max.   :61.021     Max.   :64.44
 NA's   :24             NA's   :22         NA's   :24
 lodgingscore_10flat_0erect    notes           moisture_perten5200
 Min.   : 3.000             Length:44          Min.   : 9.90
 1st Qu.: 9.000             Class :character   1st Qu.:10.00
 Median :10.000             Mode  :character   Median :10.30
 Mean   : 8.818                                Mean   :10.47
 3rd Qu.:10.000                                3rd Qu.:10.62
 Max.   :10.000                                Max.   :12.50
 NA's   :22                                    NA's   :28
 testweight_perten5200
 Min.   :15.60
 1st Qu.:27.88
 Median :31.65
 Mean   :30.76
 3rd Qu.:36.42
 Max.   :39.90
 NA's   :24
```

3 Statistics

3.1 Using combine weights and assumed TW of 32 lbs bu

```
dat2 |>
  group_by(variety) |>
  summarise(
    mean = round(mean(yield_combine32TW_buA)),
    sd = round(mean(yield_combine32TW_buA)),
    n = n() )
```

```
# A tibble: 7 x 4
  variety   mean    sd     n
  <chr>    <dbl> <dbl> <int>
1 b           20    20    22
2 buffalo     21    21     4
3 pearl       23    23     4
4 reins       30    30     3
5 rushmore    20    20     3
6 streaker     5     5     4
7 sumo        19    19     4
```

```
library(agricolae)

model_withBorders <- aov(yield_combine32TW_buA~variety, data = dat2)

summary(model_withBorders)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
variety      6   1229  204.83   2.249 0.0598 .
Residuals   37   3370   91.07
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Fail to reject Ho that there are no differences in yield when borders are included
- there are 23 border plots, let's remove them for comparison among the n=4 variety plots

```
dat2 |>
  filter(variety!="b") -> dat3

model_noBorder <- aov(yield_combine32TW_buA~variety, data = dat3)

summary(model_noBorder)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
variety      5   1213  242.68   3.114 0.0376 *
Residuals   16   1247   77.92
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
LSD.test(model_noBorder,
         trt="variety") |>
  print()
```

```
$statistics
   MSerror Df     Mean       CV
  77.92389 16 19.13078 46.14266

$parameters
        test p.ajusted  name.t ntr alpha
  Fisher-LSD      none variety   6  0.05
```

```
$means
         yield_combine32TW_buA       std r       se       LCL      UCL
buffalo              20.598218 11.560312 4 4.413725 11.241539 29.95490
pearl                23.085980  8.063193 4 4.413725 13.729300 32.44266
reins                29.768471  2.081740 3 5.096531 18.964308 40.57263
rushmore             20.420073 11.172354 3 5.096531  9.615909 31.22424
streaker              5.010604  3.238655 4 4.413725 -4.346075 14.36728
sumo                 18.883083 10.970226 4 4.413725  9.526403 28.23976
               Min       Max       Q25      Q50       Q75
buffalo   8.391878 31.430959 11.931061 21.28502 29.952176
pearl    16.643891 34.593185 18.048058 20.55342 25.591342
reins    28.208061 32.132179 28.586617 28.96517 30.548676
rushmore  7.525266 27.205790 17.027214 26.52916 26.867476
streaker  1.370673  9.033984  3.215035  4.81888  6.614449
sumo      4.481048 30.365687 14.158375 20.34280 25.067507

$comparison
NULL

$groups
         yield_combine32TW_buA groups
reins                29.768471      a
pearl                23.085980      a
buffalo              20.598218      a
rushmore             20.420073      a
sumo                 18.883083      a
streaker              5.010604      b

attr(,"class")
[1] "group"
```

- Reject Ho there are differences among varieties

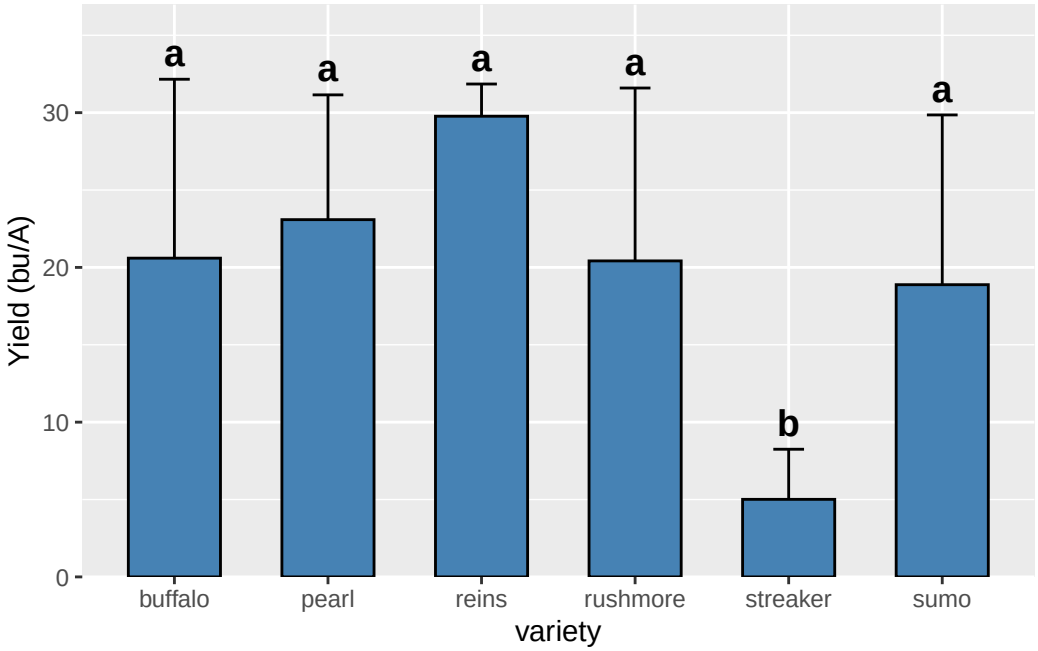


Figure 5: Oat yields compared across varieties

3.2 yield_combineAdjTW_buA

```
dat2 |>
  drop_na(yield_combineAdjTW_buA) -> dat4

dat4 |>
  group_by(variety) |>
  summarise(
    mean = round(mean(yield_combineAdjTW_buA )),
    sd = round(mean(yield_combineAdjTW_buA)),
    n = n() )
```

```
# A tibble: 6 x 4
  variety   mean    sd     n
  <chr>    <dbl> <dbl> <int>
1 buffalo     39    39     4
2 pearl       37    37     3
3 reins       50    50     3
4 rushmore    38    38     3
```

```
5 streaker    15    15     3
6 sumo        31    31     4
```

```r
model_withBorders <- aov(yield_combine32TW_buA~variety, data = dat4)

summary(model_withBorders)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
variety      5  852.9  170.57   2.269  0.104
Residuals   14 1052.6   75.18
```

```r
LSD.test(model_noBorder,
         trt="variety") |>
  print()
```

```
$statistics
   MSerror Df     Mean       CV
  77.92389 16 19.13078 46.14266

$parameters
        test p.ajusted  name.t ntr alpha
  Fisher-LSD      none variety   6  0.05

$means
         yield_combine32TW_buA       std r       se       LCL      UCL
buffalo              20.598218 11.560312 4 4.413725 11.241539 29.95490
pearl                23.085980  8.063193 4 4.413725 13.729300 32.44266
reins                29.768471  2.081740 3 5.096531 18.964308 40.57263
rushmore             20.420073 11.172354 3 5.096531  9.615909 31.22424
streaker              5.010604  3.238655 4 4.413725 -4.346075 14.36728
sumo                 18.883083 10.970226 4 4.413725  9.526403 28.23976
               Min       Max       Q25      Q50       Q75
buffalo   8.391878 31.430959 11.931061 21.28502 29.952176
pearl    16.643891 34.593185 18.048058 20.55342 25.591342
reins    28.208061 32.132179 28.586617 28.96517 30.548676
rushmore  7.525266 27.205790 17.027214 26.52916 26.867476
streaker  1.370673  9.033984  3.215035  4.81888  6.614449
sumo      4.481048 30.365687 14.158375 20.34280 25.067507

$comparison
NULL
```

```
$groups
         yield_combine32TW_buA groups
reins                29.768471      a
pearl                23.085980      a
buffalo              20.598218      a
rushmore             20.420073      a
sumo                 18.883083      a
streaker              5.010604      b

attr(,"class")
[1] "group"
```

```
## AI solution
tukey <- TukeyHSD(model_noBorder)
cld <- multcompLetters4(model_noBorder, tukey)$variety$Letters

df_summary <- dat4 %>%
  group_by(variety) %>%
  summarise(
    mean_val = mean(yield_combine32TW_buA),
    sd_val = sd(yield_combine32TW_buA)
  ) %>%
  mutate(letter = cld[variety]) # Matches letters to the correct group name

df_summary |>
  mutate(letter = c("a", "a", "a", "a", "b", "a")) -> df_summary


ggplot(df_summary, aes(x = variety, y = mean_val)) +
  geom_errorbar(aes(ymin = mean_val - sd_val, ymax = mean_val + sd_val),
                width = 0.2, color = "black") +
  # Draw the bars representing the means
  geom_col(fill = "steelblue", width = 0.6,
           col =1) +

  # Optional: Add error bars so the letters don't overlap them


  # Place the letters above the maximum height (mean + standard deviation)
  geom_text(aes(y = mean_val + sd_val, label = letter),
            vjust = -0.5,
```

```
            fontface = "bold",
            size = 5) +

  # Add 15% top padding to ensure the text doesn't hit the ceiling
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  labs(y = "Yield (bu/A)")
```

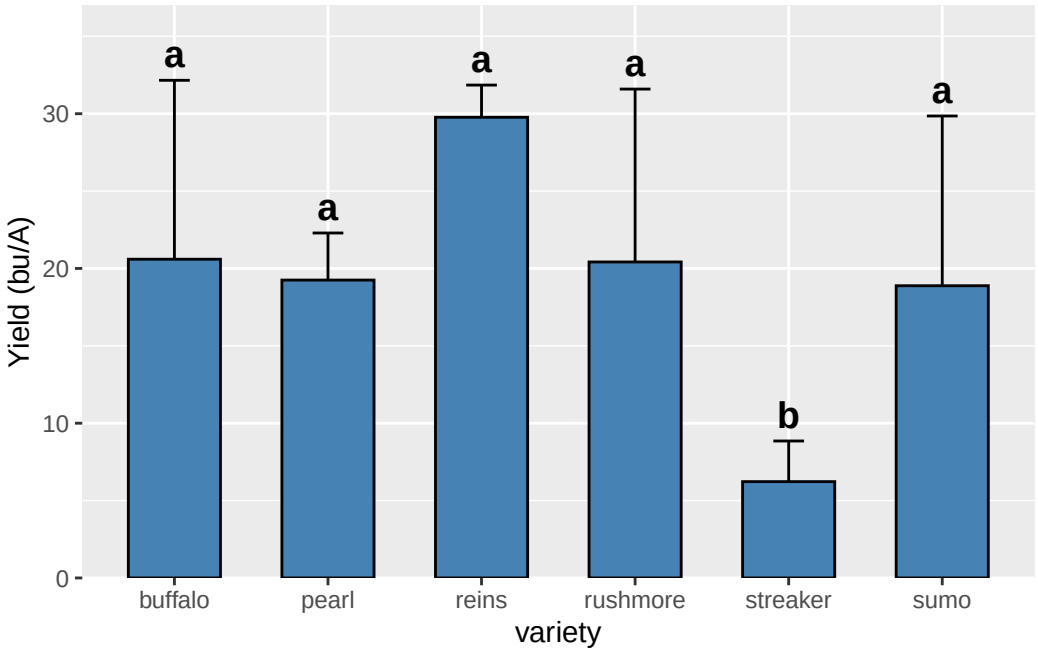


Figure 6: Oat yields compared across varieties

3.3 comparison of yield methods

```
data_long <- dat4 %>%
  pivot_longer(
    cols = c(yield_combine32TW_buA,
             yield_combineAdjTW_buA,
             yield_hand32TW_buA,
             yield_handAdjTW_buA),
    names_to = "Variable",
    values_to = "Measurement"
  )
```

```r
ggplot(data_long, aes(x = Variable, y = Measurement, fill = Variable)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) + # Hide outliers here so they aren't double
  geom_jitter(width = 0.2, alpha = 0.4, color = "black") +
  theme_minimal() +
  labs(
    title = "Comparison of Response Variables",
    x = "",
    y = "Yield (bu/A)"
  ) +
  theme(legend.position = "none",
        axis.text.x = element_text(angle=45, hjust = 1)) # Legend is redundant because x-axis
```

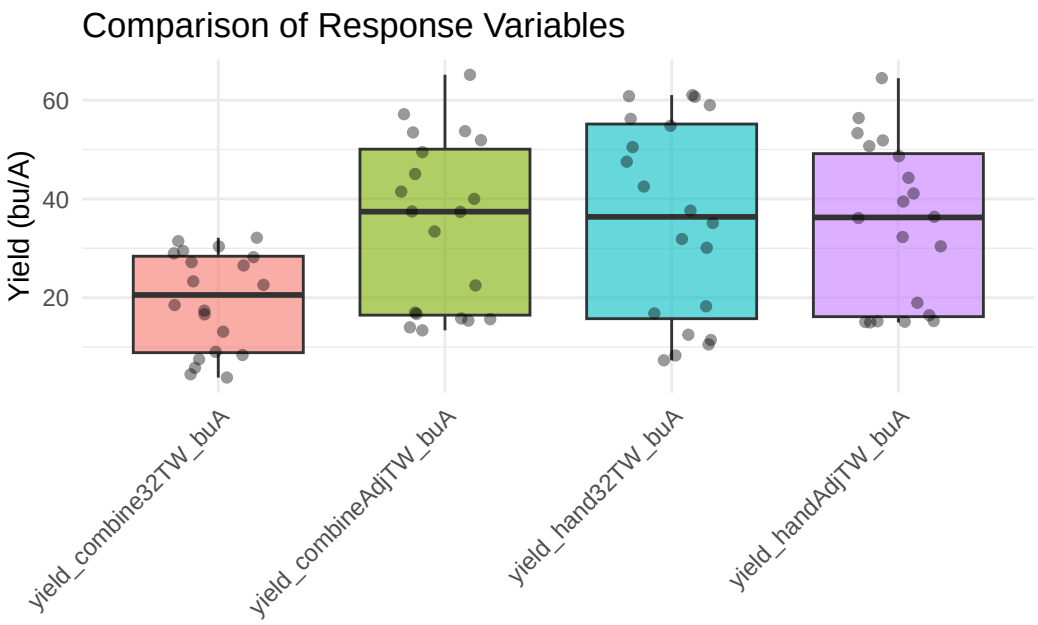
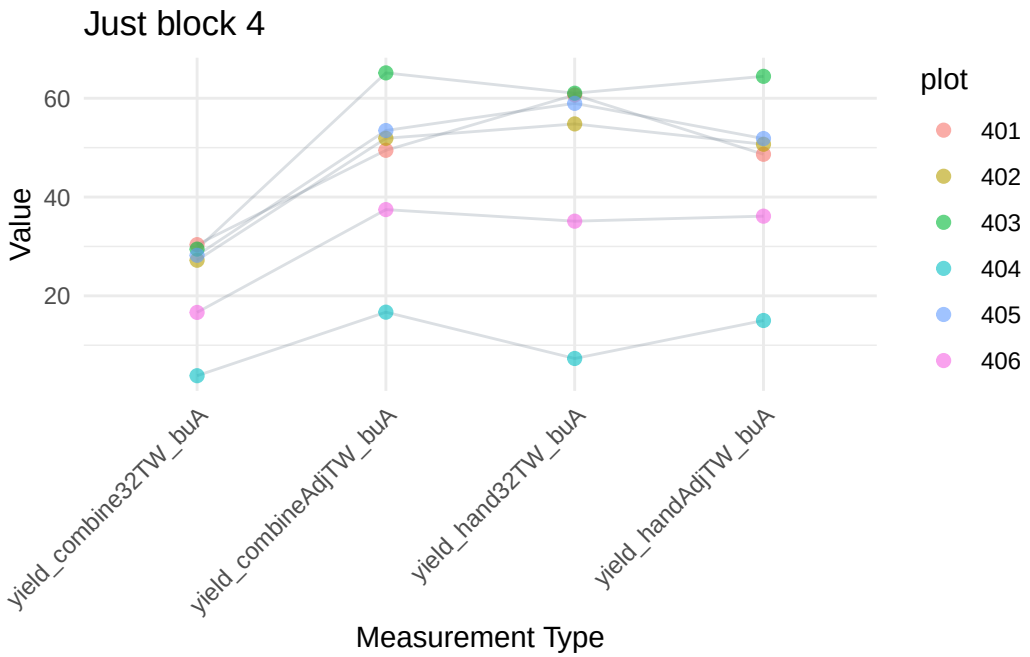


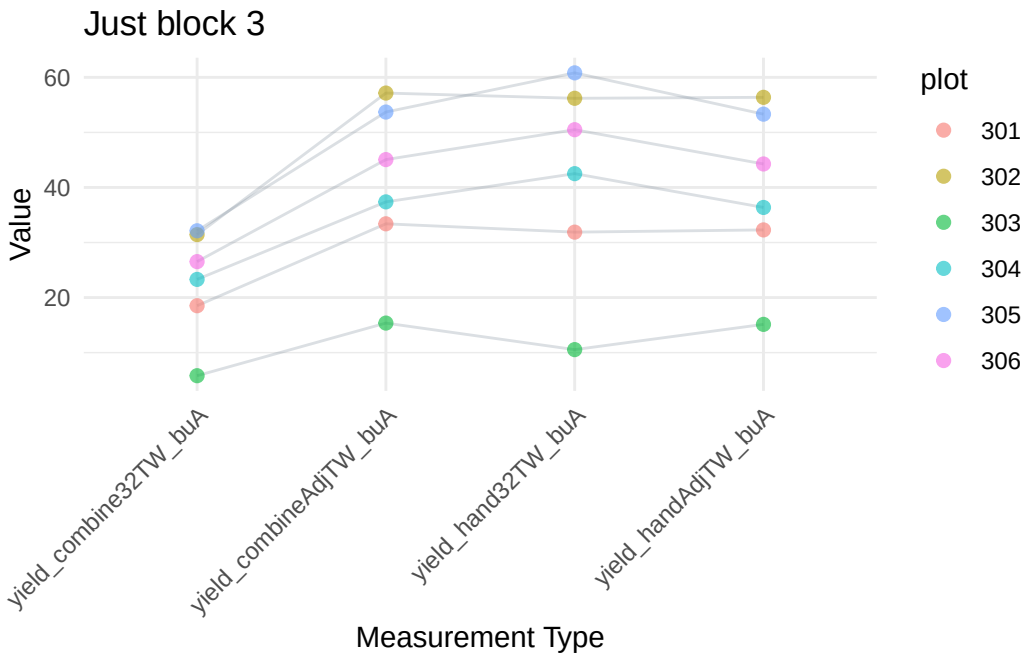
Figure 7: overall trend of yield measurements

```r
data_long |>
  filter(plot>400) |>
  ggplot(aes(x = Variable, y = Measurement, group = plot)) +
  geom_line(alpha = 0.25, color = "slategray") +       # Connects the 4 dots per ID
  geom_point(aes(color = plot), alpha = 0.6, size = 2) +
  theme_minimal() +
  labs(
    title = "Just block 4",
```

```
    x = "Measurement Type",
    y = "Value"
  ) +
  theme(axis.text.x = element_text(angle=45, hjust=1))
```




```
data_long |>
  filter(plot>300 & plot<400) |>
  ggplot(aes(x = Variable, y = Measurement, group = plot)) +
  geom_line(alpha = 0.25, color = "slategray") +       # Connects the 4 dots per ID
  geom_point(aes(color = plot), alpha = 0.6, size = 2) +
  theme_minimal() +
  labs(
    title = "Just block 3",
    x = "Measurement Type",
    y = "Value"
  ) +
  theme(axis.text.x = element_text(angle=45, hjust=1))
```


Just block 3
60
40
20
Value
yield_combine32TW_buA
yield_combineAdjTW_buA
yield_hand32TW_buA
yield_handAdjTW_buA
Measurement Type
plot
301
302
303
304
305
306